

The Cost of a Steeper Link: Calibration, Stability, and Tie Mechanics in Large-Scale Pairwise Preference Ranking

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Abstract

Large-scale preference leaderboards such as Chatbot Arena rank language models by maximum-likelihood Bradley–Terry (BT), whose logistic link assumes independence of irrelevant alternatives across a comparison pool that never stops growing. We test whether a flexible Thurstonian alternative—a lattice-based model that on pairwise data reduces to a steeper, natively tie-aware gap link—measurably improves refit stability, held-out calibration, or tie handling on 1.67 million Arena votes spanning 129 models and sixteen months. Every comparison was pre-registered: effect-size thresholds derived before fitting, synthetic detection gates passed before real data, verdict criteria fixed in advance. We find no demonstrable advantage in either direction. Refit stability under population growth is statistically indistinguishable (median paired rank-correlation difference $+0.00000$ across thirteen month-over-month refits); two structurally different tests agree that link shape is not discriminable at the ability gaps real matchups occupy (analytic ceilings of $0.23\text{--}0.31\times$ our minimum practical difference); and the calibration and tie-mechanism comparisons both end *inconclusive*, each dominated by a single-window episode rather than any systematic difference. Inside this null sit two structural findings: a steeper link’s single tie parameter entangles tie mass with decisive discrimination, whereas Davidson’s tie parameter is provably orthogonal to its logistic link; and under parameter uncertainty, steep links trade prediction-side overconfidence against an estimation-side shrinkage that once protected against a severe cold-start extrapolation error. One anomaly survives every attribution attempt: temporally localized excess dispersion under every link tested. Practically, cold-start coverage and nonstationary drift—not link choice—bind next-month predictability in deployments of this shape.

1 Introduction

Chatbot Arena [3] has become the de facto public scoreboard for large language models: users pose a prompt, two anonymous models respond, and the user votes for a winner or declares a tie. The platform converts these votes into a leaderboard by maximum-likelihood Bradley–Terry (BT) [2], fitting one latent ability per model under a logistic link. Our study window alone contains 1.8 million recorded battles across 129 models (1.67 million after the platform’s own deduplication filter), with new models entering the comparison pool continuously throughout.

BT buys its simplicity with a strong structural assumption. As a logit-family choice model it satisfies Luce’s independence of irrelevant alternatives [12]: the odds of model A beating model B are taken to be invariant to whatever else occupies the pool. Nothing about human preference data guarantees this, and a leaderboard whose pool grows by the month—forcing a global re-estimate over an ever-changing comparison graph—is precisely the setting where one might expect the assumption to bite: incumbent ratings could shift when entrants arrive, predicted win probabilities could miscalibrate for fresh models, and the tie outcome (roughly a fifth of quality votes) is not even part of BT’s native outcome space. The Thurstonian random-utility family [17] offers the classical alternative—each model draws a latent performance and the better draw wins—and Cotton’s lattice

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construction [4] makes that family computationally exact for arbitrary base densities, with ties arising natively as dead-heat mass rather than by parametric extension [5]. On pairwise data the lattice model reduces to a *gap link*: a probability curve in the ability difference, like BT’s logistic but with a different shape—steeper at a toss-up (slope 0.284–0.337 depending on its tie-band width, against the logistic’s 0.25) and carrying a native tie band. Whether that different shape is worth anything in a real deployment is an empirical question, and to our knowledge it had not been asked at this scale.

We asked it four ways, each pre-registered before any real-data number was computed: practical-significance thresholds derived from deployment-meaningful error sizes, detection gates validated on synthetic ground truth, and pattern-based verdict criteria fixed in advance. (RQ1) When new models enter, whose refit perturbs incumbent rankings less? (RQ2) Is either link shape more consistent with observed win rates across triples of models? (RQ3) Which link calibrates better on held-out votes, especially for recent entrants? (RQ4) Does a native dead-heat band fit the observed tie structure better than Davidson’s extension of BT?

The answer, delivered with unusual statistical redundancy, is that the link does not matter here. Refit stability is indistinguishable—across thirteen month-over-month refits the median paired rank-correlation difference is +0.00000, with the two methods exactly tied in five windows—while window-to-window variation in stability, governed by pool maturity, exceeds the between-method differences by roughly two orders of magnitude. Two structurally different analyses (an analytic calibration ceiling and an empirical triple-additivity test) independently agree that at the ability gaps real matchups occupy, no plausible member of the Thurstonian family is distinguishable from the logistic. The two remaining comparisons, held-out calibration and tie handling, both end *inconclusive* rather than equivalent—a distinction we maintain deliberately, because each verdict was dominated by a single anomalous window rather than by any systematic difference, and our pre-registered criteria refuse a call under exactly that pattern.

This paper’s contribution is that negative result, stated with the precision it deserves, together with what we found inside it. First, a structural asymmetry: a steep link’s single tie parameter entangles tie mass with decisive discrimination, while Davidson’s tie parameter is provably orthogonal to its logistic link—forcing the lattice band to absorb both—bad-scale tie mass measurably degrades its decisive accuracy. Second, a two-faced interaction between link steepness and parameter uncertainty: steep links make overconfident plugin predictions from noisy estimates, yet the same steepness acts as implicit shrinkage in estimation—in our data it once prevented a severe cold-start extrapolation error that cost BT dearly. Third, an open question: a temporally localized excess dispersion, present under every link we tested and attributable to none of the covariates we measured. And throughout, a deployment lesson that inverts the motivating question: what binds ranking quality in a system like Arena is not the link function at all, but cold-start coverage and nonstationary drift—both of which dwarf every between-method difference we measured. We believe the audit itself, and the pre-registration discipline that kept four successive verdicts honest, are as much the contribution as any single number.

2 Related work

Paired comparison models. The random-utility tradition begins with Thurstone [17]: each alternative draws a latent performance and the larger draw wins, giving a probit-type link under Gaussian noise [13]. Bradley and Terry [2] replaced the probit with the logistic, which Luce [12] showed is equivalent to the choice axiom now called independence of irrelevant alternatives (IIA); Harville [9] and Plackett [14] extend the same multiplicative structure to multi-entry contests and full rankings, and Davidson [5] extends BT with an explicit tie outcome. Practical rating systems descend from the same family: Elo [7] and Glicko [8] are online approximations to BT-style updating, and TrueSkill [10] is a Thurstonian (Gaussian) model fit by approximate message passing. We position TrueSkill as a relative only: it shares the Thurstonian generative story but uses approximate inference and an online update rule, and *we do not test it empirically here*—our comparison isolates the link-shape question by holding the estimation machinery fixed, which TrueSkill’s different inference style would confound. Ranking distributions without a latent performance scale (Mallows-type mod-

els and the repeated insertion family; 6) are a separate tradition; Boehmer et al. [1] warn that their behavior depends delicately on the number of alternatives, which is one reason we stay inside the random-utility family throughout.

The lattice model, and what this paper does *not* test. Cotton [4] made the Thurstonian family computationally exact for arbitrary base densities by discretizing performances onto a lattice: the distribution of the winner of an N -entrant field is computed once in $O(N)$, ties arise natively as dead-heat mass, and the construction is *coherent under field changes*—entrants can be added or removed without changing the kind of model being fit. That multi-entrant field coherence is the paper’s headline contribution, and **nothing in the present study tests it**: Chatbot Arena votes are pairwise only, and on two-entrant data the lattice construction reduces to a gap link (Section 3)—a one-dimensional probability curve in the ability difference, exactly comparable to BT’s logistic. What we evaluate is therefore the *link shape and tie mechanism* that the lattice family induces on pairwise data, with the lattice machinery serving as the computational vehicle that makes flexible Thurstonian links and exact dead-heat mass cheap to compute. A test of field coherence itself would require listwise (≥ 3 -way) preference data, which no public Arena release contains.

Arena methodology. Chatbot Arena’s leaderboard [3] is fit by maximum-likelihood BT over deduplicated anonymous battles, with ties (of both kinds) contributing half weight to each direction; our pipeline replicates the published board to a mean absolute error of 0.18 Elo points at our data snapshot (Section 4). The platform’s methodology has continued to evolve since our data window—most prominently style-controlled leaderboard variants that regress out response length and formatting; our audit uses the raw preference log that the classical leaderboard consumed, and none of our conclusions concern style adjustment.

Preference inconsistency in LLM evaluation. A recent line of work documents violations of the axioms underlying these models when LLMs are the judges or the judged: positional and permutation biases [16], instability that social-choice aggregation is proposed to repair [15], listwise preference objectives that step beyond pairwise assumptions [11], and direct measurements of transitivity and IIA violations [18]. Our triple-additivity analysis (Section 6) contributes a human-vote counterpart: a shared, temporally localized violation of *every* static gap link we test, not attributable to any measured covariate.

A structural fact we rely on. Davidson’s tie parameter is exactly orthogonal to its decisive link: conditional on a decisive outcome, the Davidson model *is* BT for every value of the tie parameter (Proposition 1). The lattice family has no such factorization—its single width parameter controls tie mass and decisive-link steepness simultaneously. This asymmetry, stated here because it is a theorem rather than an empirical finding, organizes much of what Section 8 measures.

3 Gap-link models: definitions and structure

3.1 Random utility on pairwise data

Let each model i carry a latent ability $\theta_i \in \mathbb{R}$. In a battle between A and B , each draws a performance

$$X_A = \theta_A + \varepsilon_A, \quad X_B = \theta_B + \varepsilon_B, \quad \varepsilon_A, \varepsilon_B \stackrel{\text{iid}}{\sim} p, \quad (1)$$

and the larger performance wins. Because only the difference matters, all such models are *gap links*: writing $g = \theta_A - \theta_B$,

$$\Pr(A > B) = F(g) \quad (2)$$

for some increasing F determined by the noise density p . Bradley–Terry is the special case $F = \sigma$, the logistic $\sigma(g) = (1 + e^{-g})^{-1}$, which is also the unique gap link satisfying Luce’s IIA [12].

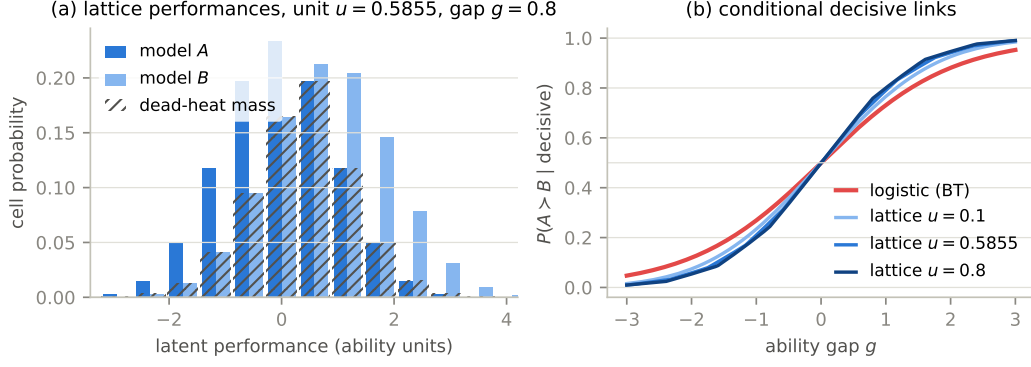


Figure 1: The lattice gap-link model. **(a)** Two models’ discretized performance distributions at gap $g = 0.8$ and cell width $u = 0.5855$ (a coarse $L = 12$ lattice is drawn so cells are visible; fits use $L = 500$). Hatched: dead-heat (tie) mass—the probability both performances land in the same cell. **(b)** Conditional decisive links (Eq. 4) for the three lattice widths used in this paper, against the logistic. The lattice links are steeper at the origin (slopes 0.290–0.337 vs. 0.25) and steepen as the tie band widens.

A continuous noise density gives ties probability zero; the two mechanisms below restore them in structurally different ways.

3.2 The lattice link: ties as dead-heat mass

Following Cotton [4], discretize the noise density onto a uniform lattice of cell width u : performances land on cells $x_j = ju$, $j \in \{-L, \dots, L\}$, with cell probabilities obtained from a base density (standard normal throughout this paper, scale 1). Writing $a_j(g) = \Pr(X_A = x_j | g)$ and $b_j = \Pr(X_B = x_j)$, the three outcome probabilities are

$$W(g) = \sum_j a_j(g) \Pr(X_B < x_j), \quad D(g) = \sum_j a_j(g) b_j, \quad L(g) = W(-g), \quad (3)$$

with $W + D + L = 1$. A tie is a *dead heat*: both performances in the same cell (Figure 1a), so the cell width u is the tie-band width— $D(0)$ grows with u —and the model has exactly one tie-related parameter, fit by profile likelihood exactly as Davidson’s ν is. The induced *conditional decisive link* is

$$F_u(g) = \frac{W(g)}{W(g) + L(g)}, \quad (4)$$

plotted in Figure 1b. Two properties matter for everything that follows. First, F_u is *steeper* than the logistic at a toss-up: $F'_u(0)$ is 0.2899, 0.3237, 0.3367 at $u \in \{0.1, 0.5855, 0.8\}$, against $\sigma'(0) = 0.25$, and it *increases* with u —carving a wider dead-heat band out of the middle of the distribution steepens what remains. (As $u \rightarrow 0$ the link approaches the probit-type limit with slope ≈ 0.284 ; the logistic’s 0.25 is unattainable at any width.) Second, and consequently, tie mass and decisive steepness are controlled by the *same* parameter: the family cannot adjust one without moving the other. We call this the *entanglement* property and measure its cost in Section 8.

3.3 Davidson: ties by extension, orthogonal by construction

Davidson [5] extends BT with a tie outcome. In gap form, with $\nu \geq 0$ and normalizer $Z_\nu(g) = 2 \cosh(g/2) + \nu$,

$$\Pr(A > B | g) = \frac{e^{g/2}}{Z_\nu(g)}, \quad \Pr(\text{tie} | g) = \frac{\nu}{Z_\nu(g)}, \quad \Pr(B > A | g) = \frac{e^{-g/2}}{Z_\nu(g)}. \quad (5)$$

Proposition 1 *For every $\nu \geq 0$, the Davidson model’s conditional decisive link is exactly logistic: $\Pr(A > B \mid \text{decisive}, g) = \sigma(g)$.*

Proof. From Eq. (5), conditioning on a decisive outcome cancels the common denominator: $e^{g/2}/(e^{g/2} + e^{-g/2}) = (1 + e^{-g})^{-1} = \sigma(g)$. $\square$

Davidson’s ν is therefore *purely* a tie-mass parameter: on decisive outcomes, Davidson and vanilla BT are the same model, and any decisive-only comparison between them is vacuous by construction. This is the exact structural opposite of the lattice’s entanglement, and it makes the two families a clean matched pair for asking whether tie mechanics have a measurable cost: both have one tie parameter and one ability per model; they differ only in *where* the tie parameter’s influence stops.

3.4 Estimation, identifiability, and reporting conventions

All fits in this paper are per-vote maximum likelihood over individual battles, by L-BFGS with analytic gradients; only the link differs between methods. Two likelihood modes are used, matching each experiment’s outcome space. The *half-tie* mode replicates the production leaderboard’s treatment [3]: each decisive vote contributes weight 2 to its direction and each tie (of either recorded kind) weight 1 to each direction, scored through the conditional decisive link,

$$\mathcal{L}_{\text{ht}}(\theta) = \sum_{\text{decisive } (w, \ell)} 2 \log F(\theta_w - \theta_\ell) + \sum_{\text{ties } (i, j)} [\log F(\theta_i - \theta_j) + \log F(\theta_j - \theta_i)]. \quad (6)$$

The *native trinomial* mode scores $\{W, D, L\}$ directly and is used where tie prediction is itself the question (Section 8). Abilities are translation-invariant, so every fit of every method anchors the same reference model, gpt-4-0613 = 0: a fixed dated checkpoint (no silent version drift), present from the third month of the data onward with high volume, hence available in every training window—the production anchor (mixtral-8x7b) enters too late for our earliest refits. Where ability *magnitudes* are compared across links (Section 5), the lattice’s are rescaled by $F'_u(0)/\sigma'(0)$ so that one reported unit buys the same win-probability change at a toss-up in every method—the exact analogue of the $400/\ln 10$ constant that turns BT coefficients into Elo points; we label such numbers convention-dependent wherever they appear. Sign conventions for the calibration deltas are stated once in each results section and held fixed.

4 Data and pre-registered methodology

4.1 Dataset and pipeline validation

We use the largest public Chatbot Arena battle log (clean_battle_20240814): 1,799,991 anonymous pairwise battles across 129 models, timestamped at sub-second resolution from 2023-04-24 to 2024-08-14, with outcomes in $\{A \text{ wins}, B \text{ wins}, \text{tie}, \text{tie (both bad)}\}$; the platform’s deduplication filter retains 1,670,250 battles, which all fits use. Ties are first-class: 20.45% of non-both-bad votes over the full period—and drifting, from 13.1% in the first three months, a fact that shapes several design decisions below.

Before any comparison, we verified the pipeline reproduces production: refitting BT at the published leaderboard’s own timestamp reproduces the published ratings to MAE 0.18 Elo points (Spearman $\rho = 0.99997$, all 128 published models). Extending the check across ten published BT-era snapshots spanning 2023-12 to 2024-08, the era-appropriate pipeline variant achieves MAE 0.18–1.01 Elo points on nine of ten; the tenth (2024-04-03) is an unexplained outlier at MAE 2.26, which we report rather than drop. As a byproduct the comparison dates the platform’s switch to deduplicated fitting to between the 2024-05-01 and 2024-06-02 snapshots.

4.2 The pre-registration protocol

Every experimental verdict in this paper was fixed before the corresponding real-data quantity was computed, in four layers.

(i) **Derived practical-significance thresholds.** Statistical significance is worthless at $n \approx 10^6$; we instead derive a *minimum practical difference* (MPD) for each comparison from a deployment-meaningful error. For decisive calibration (RQ3): a uniform 10-Elo-point error in every matchup gap—the granularity at which leaderboard consumers read ratings—is $\epsilon = 10 \ln 10/400 = 0.0576$ nats, whose expected per-vote excess log-loss at a toss-up is

$$\frac{1}{2} p(1-p) \epsilon^2 = \frac{1}{2} \cdot \frac{1}{4} \cdot (0.0576)^2 = 4.1 \times 10^{-4} \Rightarrow \text{MPD}_3 = 4 \times 10^{-4} \text{ nats/vote.} \quad (7)$$

For trinomial tie calibration (RQ4): a one-percentage-point error in $\Pr(\text{tie})$ at the observed $\sim 20\%$ tie level costs $\frac{1}{2} \Delta p^2 (1/\bar{p} + 1/(1-\bar{p})) = \frac{1}{2} (0.01)^2 (1/0.2 + 1/0.8) = 3.1 \times 10^{-4}$, so $\text{MPD}_4 = 3 \times 10^{-4}$ nats/vote. Differences below MPD are declared practically equivalent *regardless* of statistical significance.

(ii) **Effect ceilings computed before fitting.** Because both links are fit to the same votes, the largest calibration advantage either *could* have if its family were exactly true is computable in advance from the empirical gap distribution alone. With $\hat{P}$ the vote-weighted distribution of fitted gaps (median $|g| = 0.31$: real matchups concentrate near toss-ups) and F_u a lattice truth,

$$C = \min_{s>0} \mathbb{E}_{g \sim \hat{P}} \text{KL}(\text{Bern}(F_u(g)) \parallel \text{Bern}(\sigma(g/s))) \leq 0.23 \text{MPD}_3 \quad (8)$$

across every plausible width and skew, and the trinomial analogue for tie mechanisms is $\leq 0.31 \text{MPD}_4$ in *both* truth directions. These ceilings—established before any held-out vote was scored—mean the honest a-priori expectation for both calibration comparisons was practical equivalence, and any verdict exceeding a ceiling would itself demand a misspecification analysis before a mechanism claim.

(iii) **Synthetic detection gates.** The full pipeline (fitting, scoring, pooling, verdict classification) had to pass synthetic ground-truth gates before touching real outcomes: correct-sign detection with confidence intervals excluding zero at true effects down to $0.3 \times \text{MPD}$; correct positive classification of a $\geq \text{MPD}$ effect; no false positive call in any sub-MPD world; and tie-parameter recovery within 10% in rolling worlds mimicking the real entry schedule. One gate produced a finding in its own right: in a high-noise synthetic world, the *correctly specified* steep lattice loses to misspecified BT held-out ($-1.41 \times \text{MPD}$)—steep links make overconfident plugin predictions under parameter uncertainty. Grounding that noise level in the real data’s bootstrap-calibrated standard errors (Fisher information deflated by the measured factor 0.80; validated against the platform’s published bootstrap, $\rho = 0.983$) showed the reversal does *not* operate at real pooled noise—but the mechanism survives as one face of the steepness asymmetry discussed in Section 9.

(iv) **Pattern-based verdict criteria.** Verdicts are assigned by a fixed classifier over the per-window results, not by inspection. With $\bar{\Delta}$ the vote-weighted pooled mean of per-window mean deltas, CI its window-cluster bootstrap interval ($B = 10^4$; the effective sample size is the *number of windows*, 13, and is stated wherever a CI appears), and W^+ the count of windows favoring a method: *positive* requires $|\bar{\Delta}| \geq \text{MPD}$, sign consistency $W^+ \geq 10/13$ (binomial $p = 0.046$), and a CI excluding zero on the same side; *equivalence* requires $|\bar{\Delta}| < \text{MPD}$ with the CI inside $(-\text{MPD}, +\text{MPD})$ —a statistically significant but sub-MPD lean is reported as “practically equivalent despite a detectable directional lean,” never as a win; anything else is *inconclusive*, reported as such with the episode that caused it. Post-hoc analyses are permitted but labeled, answer explicitly narrower questions, and never override a pre-registered verdict.

4.3 Experimental grid

All longitudinal experiments share one grid: cumulative training windows ending at 14 monthly checkpoints (2023-07-31 through 2024-07-31, plus the final published-snapshot cutoff 2024-08-12),

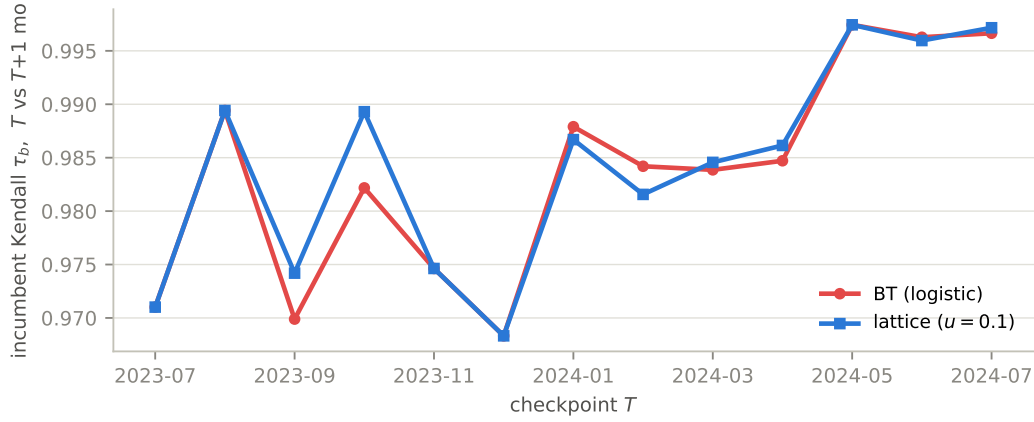


Figure 2: RQ1 headline: month-over-month incumbent ranking stability (Kendall τ_b , incumbents with ≥ 1000 votes) for BT and the lattice link across all thirteen refit windows. The two curves are statistically indistinguishable—median paired difference +0.00000, with five windows exactly tied—while both improve together as the pool matures, from $\tau_b \approx 0.97$ to ≈ 0.997 . Source: `rq1_metrics.csv`.

giving 13 train/test window pairs; testing always uses the month following the training cutoff; models absent from training are unscorable and counted (they are 24.8–75.8% of decisive test votes per window—a deployment fact we return to). Where a tie parameter is fit (Section 8), it is profiled per window on training data only. The lattice width is otherwise fixed per experiment at $u \in \{0.1, 0.5855, 0.8\}$: an arbitrary fine default; the width profiled on the *first three months only* (look-ahead-free); and the full-sample profile fit (look-ahead, retained as an upper bound). All three appear throughout because the “right” width is itself nonstationary—the drift documented in Section 8.

5 RQ1: refit stability under population growth

Question. When new models enter the pool and the leaderboard refits, whose incumbent rankings move less? For each adjacent checkpoint pair we fit both methods on cumulative data at T and $T+1$ month, restrict to incumbents (≥ 1000 votes at T ; an all-models variant is reported alongside), and compare rankings by Kendall τ_b , Spearman ρ , and maximum rank movement; ability-magnitude drift is reported in slope-matched units at horizons of one and three months, under both the common anchor and a median alignment that bounds the anchor’s own noise contribution.

Headline: a null with the signature of genuine equivalence. Across the thirteen windows the lattice is better in 5, worse in 3, and *exactly tied* in 5; the median paired τ_b difference is +0.00000, and mean τ_b is 0.98434 (lattice) versus 0.98358 (BT). The same picture holds at all three lattice widths, at both horizons, for both incumbent sets, and for every other stability metric: no incumbent moved more than five rank positions in any window under either method, and slope-matched ability drift is 1.73–1.77 Elo-equivalent points per month for every method, all within 0.04 of one another. We read the exact ties and dead-zero median as diagnostic: this is not an underpowered test failing to find a difference—at 1.67M votes the estimator differences are simply orders of magnitude smaller than the quantity being measured.

The finding that matters for practice. Figure 2 makes the secondary result visible: window-to-window variation in stability (τ_b ranging $0.968 \rightarrow 0.997$ as the pool matures) exceeds the between-method differences (≤ 0.007 , median 0) by roughly two orders of magnitude. Stability in a deployment of this shape is governed by per-model sample size and pool maturity, not by link family: an operator worried about ranking churn should buy votes for young models, not a different link. A

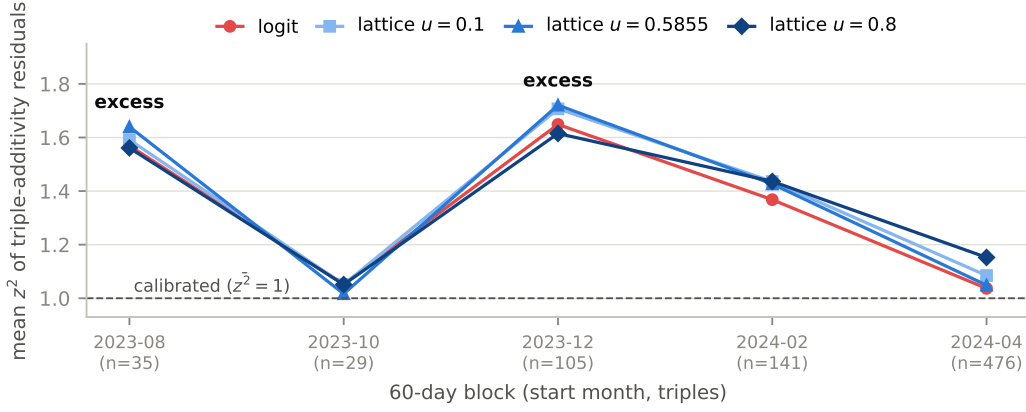


Figure 3: RQ2: mean squared additivity residual per 60-day block, per link. All four links move together—between-link differences are within the test’s synthetically measured no-power spread—while two blocks show a shared excess ($\bar{z}^2 = 1.57$ and 1.65) that no link explains. Source: `rq2b_blocks.csv`.

labeled post-hoc check found no modulation of stability—or of the (null) method difference—by entrant count or entrant vote share (all $|\rho| \leq 0.37$, descriptive $p \geq 0.22$, $n = 13$).

6 RQ2: link shape via triple additivity

Question. Any gap link makes a sharp overidentifying prediction on triples. If $\Pr(A > B \mid \text{decisive}) = F(\theta_A - \theta_B)$, then for any three models the inverted head-to-head rates must add: $F^{-1}(p_{AC}) = F^{-1}(p_{AB}) + F^{-1}(p_{BC})$. For each dense stable triple (all three legs ≥ 100 decisive votes inside a 60-day block; all three models present throughout the block) we compute the standardized additivity residual

$$z = \frac{F^{-1}(\hat{p}_{AC}) - F^{-1}(\hat{p}_{AB}) - F^{-1}(\hat{p}_{BC})}{\sqrt{v_{AB} + v_{BC} + v_{AC}}}, \quad v_{XY} = \frac{\hat{p}_{XY}(1 - \hat{p}_{XY})}{n_{XY} [F'(F^{-1}(\hat{p}_{XY}))]^2}, \quad (9)$$

by the delta method. If the link shape is right and preferences are stable within the block, $z \sim \mathcal{N}(0, 1)$ and mean $z^2 \approx 1$; excess mean z^2 measures shape misfit in a scale-invariant way, so different links compete fairly. The data admit 786 qualifying triples in five blocks. Synthetic calibration passed (true-link mean $z^2 = 1.02$ – 1.03 at matched leg sizes) and simultaneously established the test’s discrimination limit: in both generating directions the four links differ by only 0.01 – 0.04 in mean z^2 —at real gap scales this statistic *cannot* tell these links apart, which is itself the result.

Headline: indistinguishable—an independent second ceiling. Pooled mean z^2 is 1.202 (logit) versus $1.231/1.252/1.279$ (lattice widths $0.5855/0.1/0.8$), with the lattice lower in one block of five: differences of the same order as the no-power spread. This is a second, independently derived version of the effect ceiling of Eq. (8): a within-window ranking-consistency statistic with no held-out scoring, no plugin predictions, and no time splits reaches the same conclusion as the analytic KL bound—at the ability gaps real matchups occupy, no member of this Thurstonian family is distinguishable from the logistic. Two structurally different tests converging is stronger evidence than either alone.

6.1 The additivity excess: an open question

The informative structure in Figure 3 is vertical, not horizontal: in two specific eras *every* link shows the same excess dispersion. Blocks $2023-08-22 \rightarrow 10-21$ and $2023-12-20 \rightarrow 2024-02-18$ have mean

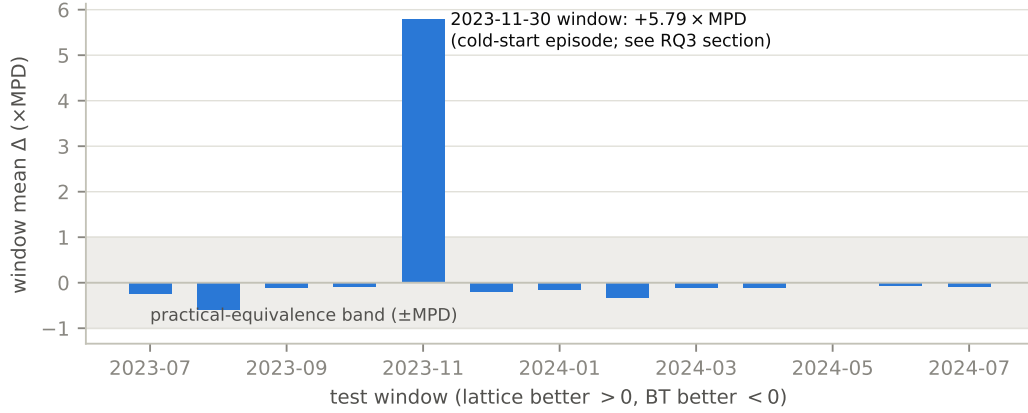


Figure 4: RQ3: per-window mean calibration delta (primary width $u = 0.5855$), in MPD units. Eleven of thirteen windows show sub-practical BT-leaning deltas inside the equivalence band; a single window carries a $+5.79 \times \text{MPD}$ excursion. The pre-registered classifier returns *inconclusive* on exactly this pattern: effect size and sign consistency disagree. Source: `rq3_window_table_u0.5855.csv`.

$z^2 = 1.57$ and 1.65 , with 13–14% of triples at $|z| > 2$ against a nominal 4.6%; the remaining blocks sit at 1.04–1.37. Half-block splits localize the first episode to roughly late September–October 2023 (half-block $z^2 = 2.04$ vs 0.91) and show the second persisting undiminished in both halves (1.74/1.54)—so within-window preference drift, which halving would attenuate, is not the mechanism. Systematic characterization ruled out, in turn: the platform’s deduplication transition (it overlaps only the *cleanest* block); entry activity (the excess blocks have *fewer* entries, 7 and 16, and lower entrant vote share, 0.62, than the clean late blocks’ 21–22 and 0.89–0.91—anti-correlated); language mix, judge counts, votes per judge, tie share, and duplicated-prompt share (none separates excess from clean blocks); and single-model artifacts (excess triples span the era’s models near base rates). Because the excess is invariant across every link tested, it is a property of the vote-generating process, not of any link choice. Two candidate explanations remain live: *judge-population composition shifts* (a mixture of judges, each individually consistent, violates additivity in aggregate) and *genuine context effects* (votes influenced by what else occupies the pool—the classical IIA-violation mechanism). Distinguishing them needs per-judge panel structure or a natural experiment this dataset does not cleanly offer; we state it as an open question, and note that an entry-based event study—the design we pre-registered for this purpose—is structurally the wrong instrument for a phenomenon anti-correlated with entries, which is why we descope rather than ran it.

7 RQ3: held-out calibration of decisive outcomes

Question and design. On next month’s decisive votes, whose predicted win probabilities are better calibrated? Both methods are fit per window in half-tie mode (Eq. (6)) and scored by log loss on the shared scoreable decisive votes. Sign convention throughout this section: $\Delta = \ell_{\text{BT}} - \ell_{\text{lattice}}$ per vote, so positive $\times \text{MPD}$ favors the lattice. Pre-registered strata: votes involving a *recent entrant* (first training appearance within 28 days of the cutoff) versus established pairs, with a noise-based pooling of windows whose recent-entrant cohorts were measurably noisy.

Pre-registered verdict: inconclusive, at all three widths. The pooled estimate at the primary width is $+0.25 \times \text{MPD}$ with CI $(-0.17, +1.22) \times \text{MPD}$ over 13 windows: the interval straddles the practical-significance boundary because 11–12 of 13 windows lean BT at sub-practical size while one window (2023-11-30) sits at $+5.8 \times \text{MPD}$ (Figure 4). The classifier refuses a call on this pattern

by design, and we report it as designed: *inconclusive is not equivalent*, and the verdict is hostage to one episode.

The episode, mechanistically. Decomposing the outlier window (labeled post-hoc): a single model, `pplx-7b-online`, entered the pool with *two* training votes at the cutoff. BT’s maximum likelihood extrapolated it to mid-table ($\hat{\theta} = +0.432$); the lattice’s steeper link—which attains the same win probabilities at smaller gaps and therefore shrinks low-data models toward the field—placed it at +0.004. The model was mass-sampled the following month and performed poorly; BT paid ≈ 68 nats over the window, $\sim 80\%$ of the window’s entire delta. This is *implicit shrinkage in estimation*: a real protective effect of link steepness, on exactly one occasion ($n = 1$; we claim the mechanism’s existence, not its frequency).

The narrower question, answered. A labeled post-hoc decomposition excludes test votes where either model has fewer than 30 training votes, asking: away from extreme cold-start, are the links equivalent? The outlier collapses ($+5.8 \rightarrow +0.04 \times \text{MPD}$) and the verdict becomes *equivalence with a sub-practical BT-leaning directional lean* at every width: pooled $-0.141 / -0.106 / -0.217 \times \text{MPD}$ at $u = 0.1 / 0.5855 / 0.8$, CIs inside the band and excluding zero. The lean is largest at the steepest width by a wide margin, directionally consistent with plugin overconfidence, but it is not monotone across the two shallower widths—suggestive, not a clean dose-response. This decomposition answers its own narrower question and does not override the pre-registered inconclusive verdict above.

The strongest single number: theory’s best-case location comes back negative. The pre-analysis identified the one place a genuine lattice-truth calibration effect could plausibly clear MPD: the recent-entrant stratum of the early high-noise windows, where under lattice-truth the predicted effect was $+1.58 \times \text{MPD}$. Measured: $-0.50 \times \text{MPD}$, CI $(-1.10, -0.06)$ —CI-bearing, and negative. In its own best-case location, the lattice-truth hypothesis is not merely unsupported but contradicted. The recent-entrant stratum overall shows the same sub-practical BT lean as everything else ($-0.12 \times \text{MPD}$, CI $(-0.18, -0.06)$).

What actually limits prediction here. Two measurements dwarf every between-method number in this section. First, reliability: fitting a recalibration slope on held-out predicted logits gives ≈ 1.455 for *all four* models—every method is underconfident on next month’s votes by the same large factor, a shared nonstationarity that no link choice addresses. Second, coverage: 24.8–75.8% of decisive test votes per window involve a model with zero training votes and are unpredictable by *any* ability model. Next-month predictability in this deployment is bound by cold-start coverage and drift, not by the link.

8 RQ4: tie mechanisms

Question and design. Does the lattice’s native dead-heat band fit observed tie structure better than Davidson’s extension, at equal parameter count? Both models are fit per window by native trinomial likelihood, abilities plus one profiled tie parameter each ($v; u$), and scored on next month’s votes over {A wins, B wins, tie}. The dead-heat analysis uses the platform’s *quality-tie* label only; “both bad” marks two responses below an absolute bar, not draws that landed close, and is analyzed separately below. Sign convention: $\Delta = \ell\ell_{\text{Davidson}} - \ell\ell_{\text{lattice}}$, positive favors the lattice; $\text{MPD}_4 = 3 \times 10^{-4}$ nats/vote.

Pre-registered verdict: inconclusive. The pattern mirrors RQ3’s, with a different cause. Pooled $+0.838 \times \text{MPD}$, CI $(+0.017, +2.542)$; the lattice is better in 10/13 windows but the estimate is carried by the same calendar window as RQ3’s outlier ($+9.55 \times \text{MPD}$). Decomposition, however, refutes the natural guess that this is the same cold-start event: the `pplx` pairs contribute -2.8% of this window’s delta. The driver is the *tie channel on near-peer, largely same-family pairs*—tie votes contribute +407 nats (lattice better) against -301 on decisive votes, led by `gpt-4-0314` vs. `gpt-4-1106-preview` and similar sibling matchups. The two pre-registered calibration comparisons were thus both driven inconclusive by single-window episodes in the same month through *different* channels—at effect ceilings of a fraction of MPD, verdicts are hostage to individual episodes, not systematic link differences. Consistently, a labeled cold-start filter that resolved RQ3’s outlier barely moves this one ($+0.861 \times \text{MPD}$, still inconclusive).

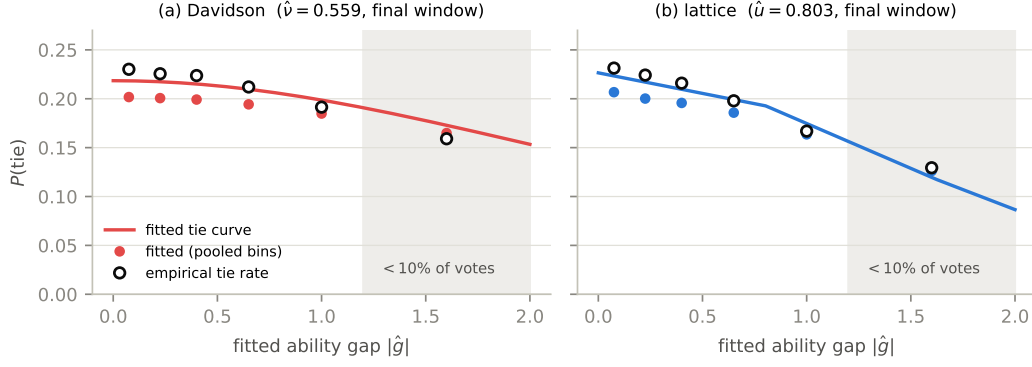


Figure 5: RQ4: fitted tie curves versus empirical tie rates, pooled over windows with vote weights, each model binned by its *own* fitted gaps. The two mechanisms agree at the origin but decay at genuinely different rates; the empirical decay lies between them, both undershoot near toss-ups, and the region where the shapes separate most ($|\hat{g}| \gtrsim 1.2$, shaded) carries under 10% of votes. Source: `rq4_tie_curve_bins.csv`.

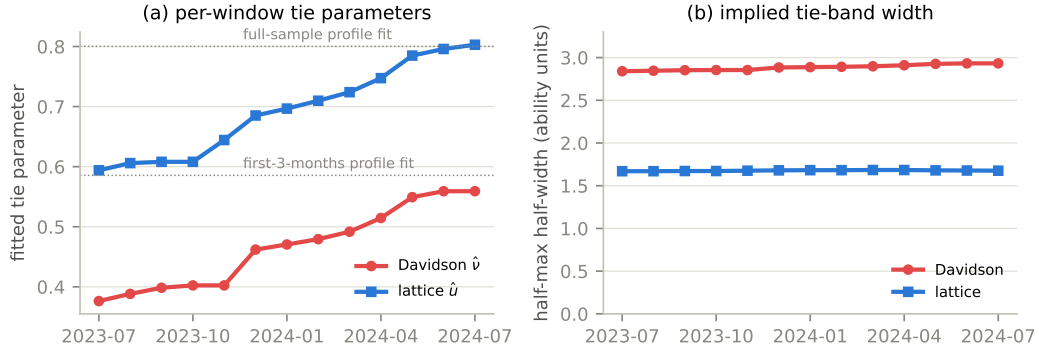


Figure 6: RQ4 drift trajectories (first-class output). **(a)** Per-window profiled tie parameters: Davidson $\hat{\nu}$ rises $0.376 \rightarrow 0.559$ and the lattice width $\hat{u}$ rises $0.594 \rightarrow 0.803$, tracking the quality-tie share ($13.1\% \rightarrow 20.45\%$). The windowed $\hat{u}$ trajectory’s endpoints independently reproduce the two earlier one-shot profile fits (dotted). **(b)** Implied half-max tie-band half-widths: the $\sim 1.7\times$ shape difference is stable even as both parameters drift. Source: `rq4_param_trajectories.csv`.

The tie-band answer: genuinely different shapes the data cannot discriminate. The two fitted mechanisms imply nearly identical at-zero tie mass in every window ($\Pr(\text{tie} \mid 0)$ within ~ 0.01) but structurally different bands: the gap at which the tie probability halves is $2.84\text{--}2.93$ ability units for Davidson’s slow sech-type decay versus $1.67\text{--}1.68$ for the lattice’s overlap decay—a stable $\sim 1.7\times$ difference across all thirteen windows (Figure 5). Yet held-out fit quality cannot separate them: vote-weighted RMS between empirical and fitted tie rates over window $\times$ gap-bin cells is 0.0225 (lattice) versus 0.0237 (Davidson), because real matchups concentrate where the shapes agree. The parameters ν and u are *not* relabelings of one quantity—they carry different shape content—but this dataset cannot say which shape is right. That is a more informative conclusion than “equivalent,” and we state it as such.

Tie propensity drifts, and both parameterizations track it. Figure 6 shows the fitted tie parameters rising near-monotonically across the study period— $\hat{\nu} : 0.376 \rightarrow 0.559$, $\hat{u} : 0.594 \rightarrow 0.803$ —mirroring the drift in raw tie share. Any single pooled tie parameter therefore averages a moving target, which is why every experiment fits per window. As an internal-validity check, the windowed $\hat{u}$ endpoints reproduce, to ~ 0.01 , two estimates computed independently months earlier by a differ-

ent procedure (a one-shot profile fit on the first three months, 0.5855, and on the full sample, 0.8002): three separately implemented estimation paths agree on the same drifting quantity.

The entanglement cost, measured. Section 3 established the structural asymmetry: Davidson’s v never touches its decisive link (Proposition 1); the lattice’s u controls tie mass and decisive steepness jointly. The both-bad robustness variant turns this into a measurement. Forcing both models to absorb both-bad-scale tie mass ($\sim 35\%$ of votes) drives the lattice width to its pre-specified profile-grid ceiling (1.4) in 11/13 windows—not for want of capacity ($u = 1.4$ expresses 40% at-zero tie mass) but because the narrow overlap band decays too fast to cover large-gap ties without inflating at-zero mass. The cost lands on the decisive channel: on identical held-out decisive votes, switching from quality-tie fits to both-bad fits degrades the lattice’s decisive log loss by $+4.75 \times \text{MPD}$ (mean over the eleven ceiling windows) versus $+2.50 \times \text{MPD}$ for the Davidson control—which isolates the shared effect of refitting abilities under the changed outcome mix, since its decisive link provably cannot move. The lattice-specific excess, $\approx +2.2 \times \text{MPD}$, is the measured price of entanglement: *in a steep-link family with one width parameter, tie mass is bought with decisive discrimination*. Qualifications: the diagnostic is modest, heterogeneous across windows, and the grid ceiling was pre-specified but never stress-tested at both-bad scale; and independently of this result, “both bad” fails the dead-heat construction conceptually—closeness of draws and a shared absolute bar are different events—which is why the category was excluded from the main analysis in the first place.

9 Discussion

9.1 One mechanism, two faces

The recurring structural actor in our results is the interaction of link steepness with parameter uncertainty, and it cuts in both directions. *Prediction-side overconfidence*: a steeper link maps ability-gap noise into larger probability errors, so plugin predictions from noisy estimates are overconfident—strong enough in high-noise synthetic worlds to make the *true* steep model lose held-out to misspecified BT ($-1.41 \times \text{MPD}$), measurably absent at our data’s pooled noise levels, and directionally visible (though not cleanly monotone) in the sub-practical BT leans of Section 7. *Estimation-side shrinkage*: the same steepness means the likelihood needs smaller gaps to explain a given record, so near-zero-data models are implicitly regularized toward the field—which prevented the cold-start episode’s ≈ 68 -nat loss exactly once in our data. Which face dominates is a property of the deployment’s noise regime, not of the link: a system with mature, well-sampled models pays the overconfidence tax for no benefit, while a system continually flooded with cold-start entrants gets shrinkage protection precisely where BT is most fragile. For reward modeling and ranking systems generally, the portable lesson is that *link steepness is an implicit uncertainty policy*: choosing a steeper-than-necessary link without uncertainty correction silently trades calibration for shrinkage, and the trade should be made deliberately—by explicit regularization or posterior-predictive prediction—rather than inherited from a link shape.

9.2 What binds ranking quality here

Every between-method difference we measured is bounded by fractions of a threshold that was itself derived from a 10-Elo reading error, while three measured quantities tower over all of them: shared underconfidence on next-month votes (recalibration slope 1.455 for every model), a quality-tie share that drifts by half again over sixteen months ($13.1\% \rightarrow 20.45\%$, dragging both tie parameters with it), and 24.8–75.8% of next-month decisive votes involving models no ability model can score. A leaderboard operator’s leverage, in order: get votes onto new models faster; recalibrate for drift; and only then worry about the link—at which point Proposition 1 argues for tie mechanisms that leave the decisive link alone.

9.3 Verdicts hostage to episodes

Both calibration comparisons ended inconclusive because of single-window episodes—different ones, through different channels (cold-start/decisive in RQ3; tie-shape/near-peer in RQ4), in the same calendar month, adjacent to one of the two unexplained additivity-excess eras. We resist connecting these dots causally; what they jointly demonstrate is a methodological point we did not anticipate when pre-registering: when true effect ceilings sit far below the practical threshold, month-scale evaluation verdicts are determined by individual episodes, and any evaluation that does not localize its verdict to its driving episodes will mistake anecdote for systematic difference—in either direction.

10 Limitations and open questions

Scope. Nothing here tests the multi-entrant field-coherence property of Cotton [4] (Section 2); our conclusions concern gap-link shape and tie mechanics on pairwise data from one platform and era. The dose-response evidence for prediction-side overconfidence is suggestive, not monotone; the shrinkage evidence is a single mechanistically explained episode. The entanglement measurement is qualified by its never-stress-tested grid boundary. One pre-registered design (an entry-proximity event study for context effects) was descoped rather than run, for the stated reason that the phenomenon it would instrument proved anti-correlated with entries; and one validation snapshot (2024-04-03, MAE 2.26) remains an unexplained pipeline outlier.

Open questions. The additivity excess (Section 6.1) is the dataset’s one genuinely unexplained signal: temporally localized, link-invariant, and unattributed after a systematic covariate sweep. Distinguishing judge-composition shifts from genuine context effects requires per-judge panel structure or listwise data—the same data that would let Cotton’s field coherence be tested directly. Both questions point at the same gap in what preference platforms currently release.

11 Conclusion

We set out to measure whether a theoretically better-motivated link function improves a real preference leaderboard, and instrumented the question carefully enough to trust the answer: it does not—not for refit stability (a dead-zero null), not for link shape (two independent tests, one ceiling), and not demonstrably for calibration or tie handling (inconclusive by episode, equivalent-with-a-lean away from episodes, and contradicted in theory’s own best-case location). The value of the exercise is what the null contains: a provable orthogonality/entanglement asymmetry between tie mechanisms with a measured cost; a two-faced steepness-uncertainty mechanism with one protective episode on record; a quantified account of what actually binds ranking quality at deployment scale; and one honest anomaly. We offer the pre-registration protocol itself—ceilings before fitting, gates before data, verdict criteria before outcomes—as the reason these negative results are informative rather than merely disappointing.

References

- [1] Niclas Boehmer, Piotr Faliszewski, and Sonja Kraiczy. Properties of the Mallows model depending on the number of alternatives: A warning for an experimentalist, 2024. arXiv:2401.14562.
- [2] Ralph Allan Bradley and Milton E. Terry. Rank analysis of incomplete block designs: I. The method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.
- [3] Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios N. Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Banghua Zhu, Michael Jordan, Joseph E. Gonzalez, and Ion Sto-

- ica. Chatbot arena: An open platform for evaluating LLMs by human preference, 2024. arXiv:2403.04132.
- [4] Peter Cotton. Inferring relative ability from winning probability in multientrant contests. *SIAM Journal on Financial Mathematics*, 12(1):295–317, 2021. doi: 10.1137/19M1276261.
 - [5] Roger R. Davidson. On extending the Bradley–Terry model to accommodate ties in paired comparison experiments. *Journal of the American Statistical Association*, 65(329):317–328, 1970.
 - [6] Jean-Paul Doignon, Aleksandar Pekeč, and Michel Regenwetter. The repeated insertion model for rankings: Missing link between two subset choice models. *Psychometrika*, 69(1):33–54, 2004. doi: 10.1007/BF02295838.
 - [7] Arpad E. Elo. *The Rating of Chessplayers, Past and Present*. Arco, New York, 1978.
 - [8] Mark E. Glickman. Parameter estimation in large dynamic paired comparison experiments. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 48(3):377–394, 1999.
 - [9] David A. Harville. Assigning probabilities to the outcomes of multi-entry competitions. *Journal of the American Statistical Association*, 68(342):312–316, 1973.
 - [10] Ralf Herbrich, Tom Minka, and Thore Graepel. TrueSkill: A Bayesian skill rating system. In *Advances in Neural Information Processing Systems 19*, pages 569–576, 2007.
 - [11] Tianqi Liu, Zhen Qin, Junru Wu, Jiaming Shen, Misha Khalman, Rishabh Joshi, Yao Zhao, Mohammad Saleh, Simon Baumgartner, Jialu Liu, Peter J. Liu, and Xuanhui Wang. LiPO: Listwise preference optimization through learning-to-rank, 2024. arXiv:2402.01878.
 - [12] R. Duncan Luce. *Individual Choice Behavior: A Theoretical Analysis*. Wiley, New York, 1959.
 - [13] Frederick Mosteller. Remarks on the method of paired comparisons: I. The least squares solution assuming equal standard deviations and equal correlations. *Psychometrika*, 16(1):3–9, 1951.
 - [14] R. L. Plackett. The analysis of permutations. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 24(2):193–202, 1975.
 - [15] Nico Potyka, Yuqicheng Zhu, Yunjie He, Evgeny Kharlamov, and Steffen Staab. Robust knowledge extraction from large language models using social choice theory, 2023. arXiv:2312.14877.
 - [16] Raphael Tang, Crystina Zhang, Xueguang Ma, Jimmy Lin, and Ferhan Ture. Found in the middle: Permutation self-consistency improves listwise ranking in large language models, 2023. arXiv:2310.07712.
 - [17] L. L. Thurstone. A law of comparative judgment. *Psychological Review*, 34(4):273–286, 1927.
 - [18] Xiutian Zhao, Ke Wang, and Wei Peng. Measuring the inconsistency of large language models in preferential ranking, 2024. arXiv:2410.08851.